

QUEENSTOWN, AB, Canada

WMO: 710730

Lat: 50.5978N

Lon: 112.8033W

Elev: 3087

StdP: 13.13

Time Zone: -7.00 (NAM)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.2	-13.9	-25.5	1.5	-18.9	-20.1	2.0	-13.6	29.6	28.0	27.4	26.9	5.7	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.1	87.3	60.3	83.7	59.1	80.1	58.2	64.3	78.8	62.2	77.1	60.5	75.9	8.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	84.3	70.2	56.6	76.6	68.7	54.4	70.6	66.1	31.1	79.0	29.5	76.7	28.2	76.2	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.7	24.1	20.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.9	18.5	15.7	26.8	39.2	51.3	58.2	64.0	62.3	54.8	40.4	29.3	16.5
	DBStd	20.64	14.05	15.84	14.83	9.79	8.75	5.87	5.32	6.39	9.50	9.78	12.07	14.62
	HDD50	5137	976	961	720	341	90	4	0	3	58	319	627	1038
	HDD65	9299	1441	1381	1185	774	430	217	83	126	324	763	1072	1503
	CDD50	1439	0	0	0	16	130	250	433	384	201	21	4	0
	CDD65	129	0	0	0	0	5	13	51	43	17	0	0	0
	CDH74	2603	0	0	0	17	177	244	929	866	363	7	0	0
	CDH80	799	0	0	0	2	33	57	294	300	112	1	0	0
(14) Wind	WSAvg	9.2	8.9	8.7	9.5	10.4	10.5	9.8	8.2	7.9	9.2	9.1	9.8	8.7
(15) Precipitation	PrecAvg	13.8	0.5	0.4	0.7	1.2	2.1	2.8	1.7	1.5	1.1	0.7	0.6	0.5
	PrecMax	21.0	1.0	0.9	2.2	2.7	3.6	7.9	4.0	3.7	4.3	1.9	1.5	1.1
	PrecMin	6.3	0.0	0.0	0.0	0.3	0.5	0.9	0.3	0.0	0.1	0.1	0.0	0.0
	PrecStd	2.6	0.2	0.2	0.4	0.6	0.9	1.5	1.0	0.9	0.9	0.4	0.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	51.3	59.5	75.9	84.2	86.5	91.6	92.6	88.7	73.1	64.4	49.0
		MCWB	37.4	39.6	43.4	50.7	54.9	61.1	61.6	60.1	58.0	52.9	46.6	36.0
	2%	DB	41.5	43.0	55.3	67.2	79.2	80.3	87.2	87.1	83.5	66.4	54.7	42.0
		MCWB	34.9	35.1	41.6	47.3	53.5	57.4	61.8	60.0	56.9	49.0	42.5	34.6
	5%	DB	38.8	39.6	50.9	61.0	74.2	76.0	83.0	83.1	78.0	61.1	49.5	38.7
		MCWB	33.6	33.0	39.4	43.7	52.1	57.2	61.1	59.2	55.2	46.6	39.6	32.8
	10%	DB	36.7	35.2	45.2	55.8	68.9	72.5	79.2	78.6	72.4	55.9	45.2	35.9
		MCWB	32.3	30.6	36.9	41.1	49.7	55.6	60.7	58.2	53.3	43.5	36.5	30.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	40.6	45.2	51.7	56.0	64.6	69.0	64.8	60.3	53.9	46.6	37.5
		MCDB	44.8	51.0	56.8	72.5	79.0	78.5	80.8	80.9	81.5	70.3	62.2	45.3
	2%	WB	35.4	35.8	42.3	47.9	54.2	61.0	65.5	62.4	58.2	50.3	42.9	35.3
		MCDB	40.6	43.0	53.6	64.3	75.6	75.0	78.2	79.9	78.2	63.3	53.4	41.6
	5%	WB	34.0	33.2	39.7	44.6	52.6	59.1	63.6	60.8	56.0	47.2	40.0	33.2
		MCDB	38.6	38.9	49.6	58.8	71.6	71.8	77.5	78.4	75.4	59.2	48.8	38.5
	10%	WB	32.4	30.7	36.7	42.1	50.7	56.9	61.8	59.2	53.9	44.6	37.2	30.8
		MCDB	36.8	34.9	44.8	54.1	66.4	69.8	75.8	75.5	70.6	54.9	44.5	35.8
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	18.2	19.6	22.7	24.3	23.3	27.1	28.0	25.5	22.4	19.0	16.4
		MCDBR	17.5	22.0	26.1	32.1	32.6	28.8	33.6	35.7	33.9	30.2	24.3	19.6
	5% WB	MCWBR	12.7	15.4	15.5	15.9	13.4	12.5	13.8	13.6	13.3	15.5	14.0	14.3
		MCWBR	17.6	20.7	24.8	30.5	29.4	25.2	27.8	31.0	31.9	28.8	24.0	19.0
(40) Clear-Sky Solar Irradiance	taub		0.257	0.264	0.284	0.333	0.352	0.345	0.346	0.362	0.320	0.281	0.254	0.236
		taud	2.418	2.474	2.475	2.341	2.340	2.406	2.424	2.358	2.453	2.537	2.498	2.435
	Ebn at Noon		254	284	296	288	286	289	287	276	278	273	256	247
		Edh at Noon	19	23	28	36	37	36	34	35	28	22	17	16
(44) All-Sky Solar Radiation	RadAvg		382	696	1116	1530	1828	1997	2170	1736	1228	756	411	287
	RadStd		29	66	83	107	128	126	107	113	115	86	38	25

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (84 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page